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DECLARATION

I Nyakundi Brian Ombati, I hereby declare that this term paper" THE RIEMANN INTEGRAL" is my original work and has not been submitted anywhere else rather than as a CAT for SMA 206 ICT class.

BRIAN NYAKUNDI OMBATI,



1/1/2024.

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ABSTRACT

Riemann integral is a mathematical concept that is defined in terms of lower and upper step functions for determining the net accumulated effect of a function over a specified interval.By dividing the interval into subintervals and approximating the function within each subinterval ,the integral captures the collective influence of the function across the entire interval.As the width of these subintervals approaches zero ,the Riemann sums converge to the definite integral,offering an exact method for calculating quantities like area and total change and this is defined as definite integral in calculus used by engineers and physicists to facilitate the analysis of dynamic processes and calculating accumulated quantities.

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CHAPTER ONE

1.**1** Notations, symbols and abbreviations

1.∑ :Summation symbol oftentimes is used when dealing with Riemann sums,which are used to approximate the Riemann integral.
2.f(x) :Denotes the function that you are integrating over a given interval.
3.dx :Indicates the variable with respect to which integration is performed.In Riemann integral,dx signifies that the variable is x.
4.[a,b] : This represents the interval over which integration is performed.The integral symbol with limits [a,b] signifies that you are integrating the function over the interval from a to b.
5.∫ :Represents integral.
6.inf and sup : Denote the infimum and supremum respectively.In Riemann integral's context they are used to find the infimum and supremum of the function values over each subinterval.
7.≤ and ≥ Denote "less than or equal to" and "greater than or equal to," respectively.Are used in this context to establish inequalities.
8. Lim :This is a limit symbol representing the limit as the width of the subintervals approaches zero, indicating the refinement of the partition and the convergence of the Riemann integral.
9. Δ. Delta notation represents width of each subinterval in the partition of the interval [a,b] used in the Riemann sum.
10.∈ denotes for member of a set
11.*P* denotes for partition
12.{} certified domain
13.[] used to show closed interval
14∀ means "for all"
15. ∃ means "existence"
16.ℕ symbol for natural numbers
17.< and >, symbol for less or greater respectively
18 := means "equivalent to"
19: $\sum_{i=1}^{n} M_i \Delta x_i$ Upper Riemann sum
20: $\sum_{i=1}^{n} m_i \Delta x_i$ Lower Riemann sum
21:μ "Mu"
22:δ "delta"
23: f^+ Right hand limit of a function
24: f^- Left hand limit of a function
25:⇒ means" implies"
26:→ means "converges to"
27:↦" maps to"

28:|f| “Absolute value of a function
29: ⇐ “conversely”
30; f “Function”
31:∞ “infinity”
32 : “ such that”
33:.dfn : means definition
34:.Thm : means theorem
35:.IVT:means Intermediate Value Theorem
36:.RS or R-Sum:stands forRiemann Sum,an approximation method
37.CDF:Cumulative Distribution Function,often used in connection with antiderivatives
38.RI or R.Int:abbreviation for Riemann Integral
39.WLOG.abbreviation for Without Loss Of Generality

1.2 DEFINITIONS

1.2.1 *Define integration.*

- This is the process of calculating the limit of a sum,where the sum involves multiplying function values by the width of subintervalsAs the width of these subintervals approaches zero,the sum converges to the definite integral,representing the area under the curve within a specified interval.

1.2.2 *Define a function.*

- A function f refers to a real-valued function defined on a closed interval [a,b].This function represents the values of a quantity,and the Riemann integral is applied to find the total accumulated “area” under the curve of f within the specified interval.The process involves partitioning the interval into subintervals,and summing the product of function values and subintervals,selecting sample points within each subinterval,and summing the product of a function values and subinterval widths as the subintervals approach zero width.

1.2.3 *Define sets.*

- In Riemann integral,sets are used to create partitions that divide the closed intervals [a,b].A partition ***P***,consists of a finite set of points $\{x_0,x_1,....,x_n\}$ such that a = x_0< x_1<....<x_n = b.These points determine subinterval.The Riemann integral is then defined as the limit of these sums as the maximum subinterval width approaches zero.

1.3 LITERATURE REVIEW

1.3.1 *Historical development of Riemann integral.*

Newton and Leibniz independently developed the concept of integration in the late 17th century.Newton used the method of fluxions,which was based on the idea of infinitesimals,while Newton used indivisibles method, which was based on the idea of dividing a curve into infinitely many small parts.After then Riemann introduced the Riemann integral in his 1854 paper “on the representation of a function,by a Trigonometric series.”He defined the integral of a function over an interval as the limit of the sum of the areas of the rectangle under the graph of the function,a method known as Riemann sums.

The Riemann integral was further developed by Henry Lebesgue integral,which is a more general concept of integration that extends the Riemann integral.The Lebesgue integral is based on the concept of measure,and it allows for the integration of a wider class of functions than the Riemann integral.

1.3.2 *Importance of Riemann integral and people who improved it*.

The Riemann integral is a method of defining the integral of a function over an interval.It was introduced by the German mathematician Bernhard Riemann in the 19th century.The Riemann integral is based on the concept of dividing the interval into a finite number of subintervals,approximating the function on each subinterval,and then taking the limit as the number of subintervals approaches infinity.
It's also a fundamental concept in calculus and mathematical analysis.It provides a way to calculate the area under a curve,which is a key problem in calculus.The Riemann integral is also used to define the concept of a continuous function,which is a fundamental concept in analysis.

1.3.3 *People who have improved Riemann integral*:

-**Henry lebesgue**:He developed the Lebesgue integral,which is a generalization of the Riemann integral and is used in modern analysis

-**Bernhard Riemann:**Introduced Riemann integral concept in 1854 paper “On the representability of functions by Trigonometric series”.

-**Karl Weierstrass:**Provided a rigorous definition of the Riemann integral in terms of limits of Riemann integral.

-**Augustin-Louis Cauchy:**Made vital contributions to the development of the integral calculus,including a precursor definition of the Riemann integral in terms of limits of Rieman sums.

CHAPTER TWO

2.1 Integrable functions Discussions:

2.1.1 *Riemann integral of a function.*

-This is a number that represents the area under the graph of a function

2.1.2 *Integrable function.*

-This is a function that can be approximated by simple functions,where a simple function is a finite linear combination of indicator functions of intervals

2.1.3 *Simple function.*

-This is a function that takes on a finite number of values.

2.1.4 *Indicator function.*

-This is a function that takes on the value 1 within the interval and 0 outside the interval

2.1.5 *Approximating a function by simple functions.*

-This function means expressing the function as a limit of a sequence of simple functions

2.2 Theorem 1

A function f(x) is said to be Riemann integrable on [a,b] if,for any given $\in > 0$,there exists a partition $\boldsymbol{P}$ of the interval [a,b] such that the Riemann sum S($\boldsymbol{P}$,f) satisfies

$$\left| S(\boldsymbol{P}, f) - \int_a^b f(x)dx \right| < \in$$

Here, $\boldsymbol{P}$ is a partition of the interval [a,b],and the Riemann sum S($\boldsymbol{P}$,f) is given by:

$$S(\boldsymbol{P},f) = \sum_{i=1}^{n} f(c_i)\Delta x_i$$

where c_i is any point in yeh ith subinterval of the partition $\boldsymbol{P}$, and Δx_i is the width of the ith subinterval.

If a function satisfies this condition for all possible partitions and corresponding Riemann sums,then the Riemann integral of f(x) over [a,b] exists.

2.3 Theorem 2

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded function defined on the closed interval [a,b].Then, f is Riemann integrable on [a,b] if and only if,for every $\epsilon > 0$, $\exists\, \boldsymbol{P}$ of [a,b] : U($\boldsymbol{P}$,f) - L($\boldsymbol{P}$,f) < $\in$

Where,

U($\boldsymbol{P}$,f) is the upper limit sum f over the partition $\boldsymbol{P}$,

L($\boldsymbol{P}$,f) is the lower limit sum f over the partition $\boldsymbol{P}$.

f is integrable if,for any small positive number $\in$,there exists a way to partition the interval [a,b] such that the difference between the upper and lower sums is arbitrarily small.

2.3 Theorem 3

Let f : [a,b] → $\mathbb{R}$ be a simple function defined on the closed interval[a,b] with values $c_1, c_2 \ldots, c_n$ .Then, f is Riemann integrable on [a,b] and its integral over[a,b] is given by:

$$\int_a^b f(x)dx = \sum_{i=1}^{n} c_i \cdot \text{length}(E_i)$$

where E_i is the set of a point in [a,b] where f takes the value c_i,and length (E_i) is the length of the interval over which f is constant at c_i.

2.4 Theorem 4

Let f :[a,b] → $\mathbb{R}$ to be Riemann integrable function on a closed interval [a,b].Then,for any ∈ > 0,there exists a simple function ϕ : [a,b] → $\mathbb{R}$ such that

$$\int_a^b |f(x) - \phi(x)|dx < \in$$

2.5 Theorem 5

Let f :[a,b] → $\mathbb{R}$ to be Riemann integrable function on a closed interval [a,b].Then,for any ∈ > 0,there exists a step function ϕ : [a,b] → $\mathbb{R}$ such that

$$\int_a^b |f(x) - \phi(x)|dx < \in$$

2.5.1 *Example for Riemann integral of a function*

Let ƒ (x) = x - 1 for x in [0, 2] and f(x) = 3 - x for x in (2, 4]. Then, | ƒ (x)| = | x - 1| for x in [0, 2] and | ƒ (x)| = |3 -x | for x in (2, 4].

Since both |x- 1| and |3 - x| are continuous on their respective intervals, they are Riemann integrable on [0, 2] and (2, 4], respectively. Therefore, | $f(x)$| is Riemann integrable on [0, 4]

The Riemann integral of |f(x)| on [0, 4] can be calculated as follows:

$$\int_0^4 |f(x)|dx = \int_0^2 |x-1|dx + \int_2^4 |3-x|dx$$

$$= \int_0^1 (1-x)dx + \int_1^2 (x-1)dx + \int_2^3 (x-3)dx + \int_3^4 (3-x)dx$$

=2

Therefore, the Riemann integral of | $f(x)$| on [0, 4] is equal to 2.

2.5.2 *Example for integrable function*

Let $f(x)= \frac{1}{x}$ for in x [1,2].We can partition the interval [1,2] into n subintervals of equal width $\Delta x = \frac{(2-1)}{n}$ .Let xi be the right end point of the i-th subinterval,then xi =1 + iΔx.The Riemann sum of $f(x)$ on [1,2] w.r.t this partition is:

$$\sum_{i=1}^{n} f(x_i)\Delta x_i = \sum_{i=1}^{n} \frac{1}{1+i\Delta x} \Delta x$$

As n approaches infinity then:

$$\int_1^2 \frac{1}{x} dx = \ln(2)$$

Therefore the Riemann integral of f(x) on [1,2] is equal to $\ln(2)$

2.5.3 *Example for simple function*

simple function of Riemann integral for $f(x)$= 2x on the interval [0,2] is

$$f(x)= \sum_{i=1}^{4} i x_{[\frac{i-1}{2}, \frac{i}{2}]}(x)$$

where $x_{[a,b]}(x)$ is the characteristic function of the interval [a,b], which is 1 if $x\in$[a,b] and 0 otherwise.

This expression is obtained by dividing the interval [0,2] into 4 equal subintervals and assigning the value of f at the right endpoint of each subinterval to the whole subinterval.

The Riemann integral of this simple function on [0,2] is

$$\int_0^2 f(x)dx = \sum_{i=1}^{4} i.\frac{1}{2} = \frac{1}{2}\sum_{i=1}^{4} i = \frac{1}{2}.10 = 5$$

2.5.3 *Example for indicator function*

Consider the indicator function of the interval [0,1]. The Riemann sum of this function over the interval [0,1] can be calculated as follows:

$$\sum_{i=1}^{n} f(x_i)\Delta x_i = \sum_{i=1}^{n} [0,1]\Delta x_i$$

where Δx_i the width of the i-th subinterval and is x_i a point in the i-th subinterval.

Since the function is constant over each subinterval, we can choose any point in the subinterval to evaluate the function. Let's choose the right endpoint of each subinterval. Then we have:

$$\sum_{i=1}^{n} f(x_i)\Delta x_i = \sum_{i=1}^{n} [0,1]\Delta x_i = \sum_{i=1}^{n} \Delta x_i$$

Since the function is the indicator function of the interval [0,1], it takes the value 1 on the interval [0,1] and 0 elsewhere. Therefore, the Riemann sum is equal to the sum of the widths of the subintervals, which is equal to the length of the interval [0,1]. Hence, the Riemann integral of the function $f(x)$ = 1[0,1] over the interval [0,1] is:

$$\int_0^1 [0,1]dx = 1-0 = 1$$

2.5.4 *Approximating a function by simple functions.*

Let $f(x) = x^3$ on the interval [0, 1]. We can use the following steps to approximate f by simple functions of Riemann integral:

Divide the interval [0, 1] into n subintervals of equal length Δx = 1/n.

Define the step function g(x) as follows:

∀ k = 0, 1, 2, ..., n-1, let g(kΔx) = f(kΔx).

∀ k = 1, 2, ..., n-1, let g(x) = f(kΔx) on the interval (kΔx, (k+1)Δx].

Let g(1) = f (1).

Define the continuous function h(x) as follows:

∀ k = 0, 1, 2, ..., n-1, let h(x) = g(kΔx) on the interval [kΔx, (k+1/2)Δx].

∀ k = 0, 1, 2, ..., n-1, let h(x) = g((k+1)Δx) on the interval [(k+1/2)Δx, (k+1)Δ]x.

Then, we have g(x) ≤ f (x) ≤ h(x) for all ix n [0, 1].

In conclusion, we can calculate the Riemann integral of |g(x) - h(x)| over [0, 1] to get an estimate of the error in our approximation.

3.1 Non integrable functions.

3.1.1 *Hyperbolic functions:*

These are functions that are similar to trigonometric functions,but they are defined for the hyperbola instead of the circle.

3.1.2 *Rational functions:*

These are functions that can be expressed as the quotient or fraction of two polynomials.

3.1.3 *Trigonometric functions:*

These are functions that relate the angles of a triangle to the lengths of its sides.

3.1.4 *Exponential functions:*

These are functions in the form of $f(x) = a^x$,where

3.1.5 *Logarithmic functions:*

These are the functions in the form of $f(x) = log_a(x)$.Where a is a constant and x is variable.

3.2 Theorem 1

The basic hyperbolic functions are:

Hyperbolic sine “sinh” (sinh)

Hyperbolic cosine “cosh” (cosh)

Hyperbolic tangent “tanh” (tanh)

Hyperbolic cotangent “coth” (coth)

Hyperbolic secant “sech” ($\sech

3.3 Theorem 2

A rational function is a function which is the ratio of polynomial functions. Said differently, r is a rational function if it is of the form r(x)=$\frac{p(x)}{q(x)}$,where p and q are polynomial functions.

3.4 Theorem 3

Pythagorean Theorem: In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides. This theorem is often written as a^2+b^2 +c^2 where a and b are the lengths of the legs of the triangle, and c is the length of the hypotenuse

3.5 Theorem 4

If $f(x) = a^x$and g(x) = b^x are exponential functions with positive bases a and b, then:

$f(x)$ = g(x) if and only if a = b or x= 0

$f(x)$ < g(x) if and only if 0 < a < b and > 0, or 0 < b < a and x< 0

3.6 Theorem 5

If $f(x) = log_b x$ and g(x) = $log_c x$ are logarithmic functions with positive bases b and c, then:

$f(x)$ = g(x) if and only if b = c or x= 1

$f(x)$ < g(x) if and only if 0 < b < c and x > 1, or 0 < c < b and x< 1

3.6.1 *Example for hyperbolic function*

The shape of a hanging cable, such as a power line, is called a catenary. It can be described by the equation y = a cosh ($\frac{x}{a}$), where a is a positive constant that depends on the weight and tension of the cable. The graph function is shown below:

y=a cosh($\frac{x}{a}$)

Some properties of the catenary are:

The lowest point of the curve is at the origin, where y = a.

The curve is symmetric about the y-axis.

The curve approaches the lines y = ±x as $\rightarrow x$ ±∞.

3.6.2 *Example for rational functions*

g(x) = $\frac{x^2-4}{x^2-9}$To graph this function, we need to identify its vertical asymptotes, horizontal asymptotes, and holes. A vertical asymptote is a vertical line where the function is undefined, a horizontal asymptote is a horizontal line where the function approaches as x goes to infinity, and a hole is a point where the function is undefined but the limit exists.

For the given function, we can factor the numerator and denominator as follows:

g(x) = $\frac{(x-2)(x+2)}{(x-3)(x+3)}$

We can see that there are no common factors in both the numerator and denominator. This means that the function has no holes.

To find the vertical asymptotes, we set the denominator equal to zero and solve for x:

x= ±3

So, the vertical asymptotes are x= 3 and x= -3.

To find the horizontal asymptote, we compare the degrees of the numerator and denominator. Since they are equal, the horizontal asymptote is the ratio of the leading coefficients:

y = 1 / 1

y = 1

So, the horizontal asymptote is y = 1.

Using these information, we can sketch the graph of the rational function as shown below:

g(x)=$\frac{x^2-4}{x^2-9}$

3.6.3 *Example for trigonometric functions*

Find the length of a side of a triangle. Suppose we know that the triangle has a right angle, an angle of 60 degrees, and a hypotenuse of 10 cm. We can use the cosine function to find the length of the adjacent side,

The cosine function is defined as the ratio of the adjacent side to the hypotenuse of a right triangle. In this case, the adjacent side is the length we want to find, which we denote by x, and the hypotenuse is 10 cm. Therefore,

cos 60° = $\frac{x}{10}$

Using the trigonometric table, we can find the value of cos 60°:

cos 60° = $\frac{1}{2}$

Substituting this value into the equation, we get:

$\frac{1}{2}$ = $\frac{x}{10}$

Multiplying both sides by 10, we get:

x = 5

Therefore, the length of the adjacent side is 5 cm.

3.6.4 *Example for exponential functions*

Let $f(x)$ = $5(3)^{x+1}$.Evaluate f(2).

f(2) = $5(3)^{2+1}$

= $5(3)^{2+1}$

=$5(27)$

=135

3.6.5 *Example for logarithmic functions*

Let $f(x)$ = ln(x) be defined on the interval [1, e]. We will show that f is Riemann integrable on [1, e].

Let P = {1, $e^{\frac{1}{n}}$, $e^{\frac{2}{n}}$, …, $e^{\frac{n-1}{n}}$, e} be a partition of [1, e] into n subintervals of equal length. Then the length of each subinterval is $\Delta x = \frac{e-1}{n}$. Let xi be any point in the ith subinterval. Then xi = $e^{\frac{i-1}{n}}$ for i = 1, 2, …, n.

The Riemann sum for f with respect to P is

S(P,f)= $\sum_{i=1}^{n} f(x_i)\Delta x_i = \sum_{i=1}^{n} \ln e^{\frac{i-1}{n}} . \frac{e-1}{n} = \frac{e-1}{n} \sum_{i=1}^{n} \ln e^{\frac{i-1}{n}}$

$= e^{\frac{n-1}{n}} \sum_{i=1}^{n} \frac{i-1}{n} = \frac{e-1^{n-1}}{n^2} \sum_{i=0}^{n-1} i = \frac{e-1}{n^2} . \frac{(n-1)n}{2} \frac{e-1}{2n}$.(e-1)Since f is continuous on [1, e], it follows that f is Riemann integrable on [1, e]. The Riemann integral of f over [1, e] is given by $\int_1^e \ln(x) dx = \lim_{n \to \infty} S(P,f)$

$\lim_{n \to \infty} \frac{e-1}{2n}$.(n-1)=e-1.Therefore, the Riemann integral of ln(x) over [1, e]

CHAPTER THREE

4.1 Riemann integral

4.1.1 Riemann Integral Definition

Let us suppose that f is a function, defined over a closed interval represented by [a, b]. Suppose f is a non-negative and continuous function. The integral of f with respect to x signifies the area between the graph of f and X-axis. This area would be called the definite integral of a function f from a to b.

4.1.2 Riemann Integral

The Riemannian method of determining this area (or definite integral) is: “if the area is divided into rectangles of very small breadth, then it would be simpler to calculate the area of such rectangles and then we can add them up.

4.1.3 Riemann method

It is quite apparent that if these rectangles are very narrow, then the approximation error would be negligibly small. For the narrower rectangles, the measured approximated area may converge to a number which is called the area under the curve of the function f. If we do not draw tiny rectangles, then this method tends to break down. To make the Riemann procedure accurate, we may determine the widths of rectangles by splitting the given interval [a, b] into smaller parts.

4.1.4 Concepts in Riemann integral

i)Partition:

Suppose a closed interval [a, b] ∈ R.

The partition of this interval is said to be a sequence of the following form:

a = $x_0 < x_1 < x_2 < ... < x_n$

where, every $[x_i, x_{i+1}]$ is known as a sub-interval.

ii)Norm:
The length of the biggest sub-interval is called a norm or a mesh, i.e. norm = max

iii)Tagged Partition:
For interval [a, b], a partition P(x,t) along with a sequence of finite numbers $t_0, t_1, ..., t_{n-1}$ is known as tagged partition if it satisfies the condition that, $t_i \in [x_i, x_{i+1}]$ for every i.

4.1.5 Riemann Sum:

The Riemann sum of a real-valued function f on the interval [a, b] is defined as the sum of f with respect to the tagged partition of [a, b]. I.e.

Every term in Riemann sum denotes the area of a rectangle having length or height $f(t_i)$ and breadth x_i+1-x_i. Hence, the Riemann sum gives the area of all the rectangles and thus the area under the curve within the interval [a, b] or definite integral.

4.1.6 Riemann Integral Formula

Let f be a real-valued function over the interval [a, b] and let L be a real number. Then, f is called integrable within [a, b] if and only if there is a δ > 0 for each ϵ > 0 such that for each partition having a property that ||P|| < δ, we can have;

$|S(f,P)-L| < \epsilon$

Where, L is known as the integral of f over the interval [a, b], thus we write it as below :

$$L = \int_b^a f(x)\, dx$$

4.1.7 Properties of Riemann Integral:

i)Linearity

ii)Monotonicity

iii)Additivity

Linearity

If f: [a, b] → R is integrable and c ∈ R, then cf is integrable and

$$\int_a^b cf = c\int_a^b f$$

And also,

$$\int_a^b (f+g) = \int_a^b f + \int_a^b g$$

Monotonicity

As per this property, if $f \leq g$, then;

$\int_a^b f \leq \int_a^b g$

Additivity

If a < c < b, then;

$\int_a^c f + \int_c^b f = \int_a^b f$

5.0 Theorems

5.1 Theorem 1

If $f(x)$ is monotone on a closed interval [a,b],then $f(x)$ is Riemann integrable on [a,b].

5.2 Theorem 2

If $f(x)$ and g(x) are Riemann integrable on [a,b],then $f(x)$ + g(x) is also Riemann integrable on [a,b] and $\int_a^b (f(x) + g(x))dx = \int_a^b f(x)dx + \int_a^b g(x)dx$.

5.3 Theorem 3

Let f be a bounded real-valued function on [a,b].

Then: $\int_{-a}^{b} f \leq \int_{a}^{-b} f$

5.4 Theorem 4

Let f be a bounded function on[a,b].If f is Riemann integrable then $|f|$ is also Riemann integral.

5.5 Theorem 5

Let f be a bounded function on [a,b].If f is continuous then f is Riemann integrable.

5.1.1 *proof 1*

Let P be a partition of [a, b] and μ(P) be the mesh of P. Then since f is monotonic increasing, we have,

$M_i = f(x_i)$ and $m_i = f(x_{i-1})$

Thus: U(P, f) − L(P,f) = $\sum_{i=1}^{n} M_i \Delta x_i - \sum_{i=1}^{n} m_i \Delta x_i$

$= \sum_{i=1}^{n} (M_i - m_i)\Delta x_i$

$= \sum_{i=1}^{n} ((f(x_i) - f(x_{i-1}))\Delta x_i$

$\leq \mu(P) \sum_{i=1}^{n} ((f(x_i) - f(x_{i-1})) = \mu(P)(f(b) - f(a)$

But for each $\epsilon > 0$, P can be chosen so that:

$\mu(P) < \frac{\epsilon}{f(b) - f(a)}$

Hence we have:

U(P,f) − L(P, f) < μ(P)(f(b) - f(a) < $\frac{\epsilon(f(b)-f(a))}{f(b)-f(a)}$ = ϵ

∴ U(P,f) − L(P, f) < ϵ

Hence f is Riemann integrable.

5.2.1 *Proof 2*

Let f: [a, b] → ℝ and g: [a, b] →ℝ be Riemann integrable functions. We want to show that f+ g is also Riemann integrable.

Let P = {x0, x1, …, xn} be a partition of [a, b]. Then, we have:

L(f, P) + L(g, *P*) ≤ L(f+ g, *P*) ≤ U(f + g, *P*) ≤ U(f, *P*) + U(g, *P*)

where L(f, *P*) and U(f,*P*) are the lower and upper Riemann sums of f over *P*, respectively, and similarly for g and f+ g.

Since f and g are Riemann integrable, we know that their upper and lower Riemann sums converge to the same value as the mesh size of the partition goes to zero. Therefore, we have:

$\lim_{T}(\text{mesh}(P)\to 0)\ \{L(f, P) + L(g, P)\} = \int_{-a}^{b} f(x)\,dx + \int_{-a}^{b} g(x)\,dx$

and

$\lim_{T}(\text{mesh}(P)\to 0)\ \{U(f, P) + U(g, P)\} = \int_{-a}^{b} f(x)\,dx + g(x)\int_{-a}^{b} dx$

$\therefore \int_{-a}^{b} (f(x) + g(x))\,dx = \lim_{T}(\text{mesh}(g)\to 0)\ \{L(f + g, P)\} = \lim_{T}(\text{mesh}(P)\to 0)\ \{U(f + g, P)\}$

5.3.1 *Proof 3*

(⇒) Let f be Riemann integrable over [a, b]. Then

$\int_{a}^{-b} f\,dx = \int_{-a}^{b} f\,dx = \int_{a}^{b} f\,dx$

But

$\int_{a}^{-b} f\,dx$ = inf { U(*P*, f) : *P* ∈ ℙ [a,b]}

$\int_{-a}^{b} f\,dx = \sup\{ L(P,f) : P \in \mathbb{P}[a, b]\}$

Therefore, given any $\in > 0$, $\exists$ $P1, P2 \in \mathbb{P}[a, b]$ such that:

$U(P1, f) < \int_{a}^{-b} f\,dx + \frac{\epsilon}{2} = \int_{a}^{b} f - \frac{\epsilon}{2}$

And

$L(P2, f) > \int_{-a}^{b} f\,dx - \frac{\epsilon}{2} = \int_{a}^{b} f - \frac{\epsilon}{2}$

Now, let $P = P_1 \cup P_2$. Therefore $P_1 \subseteq P$ and $P_2 \subseteq P$

Hence

$$U(P,f) < \int_{a}^{b} f\,dx + \frac{\epsilon}{2}$$

and $$U(P, f) < \int_{a}^{b} f\,dx - \frac{\epsilon}{2}$$

Thus

$\int_{a}^{b} f\,dx - \frac{\epsilon}{2} L(P,f) < L(P, f) < U(P, f) < \int_{a}^{b} f\,dx + \frac{\epsilon}{2}$

That is:

$U(P,f) - L(P,f) < (\int_{a}^{b} f\,dx + \frac{\epsilon}{2}) - (\int_{a}^{b} f\,dx - \frac{\epsilon}{2}) < \int_{a}^{b} f\,dx + \frac{\epsilon}{2} - \int_{a}^{b} f\,dx + \frac{\epsilon}{2} = \epsilon$

That is

$$U(P, f) - L(P, f) < \epsilon$$

($\Leftarrow$) Conversely, for each $\epsilon > 0$, let $P \in \mathbb{P}[a, b]$ such that

$U(P,f) - L(P,f) < \epsilon$

Then we have

$$U(P,f) \geq \int_{a}^{-b} f\,dx \geq \int_{-a}^{b} f\,dx \geq U(P,f)$$

Therefore

$$\int_{a}^{-b} f\,dx \geq \int_{-a}^{b} f\,dx = U(P,f) - L(P,f) < \epsilon, \ \forall\ \epsilon > 0.$$

Since $\epsilon > 0$ is arbitrary, we have that

$$\int_{-a}^{b} f\,dx = \int_{a}^{-b} f\,dx = \int_{a}^{b} f\,dx$$

Hence f is Riemann integrable.

5.3.1 *Proof 3*

Given any two partitions P_1 and of P_2 [a, b],

$L(P_1, f) \leq L(P_1 \cup P_2, f) \leq U(P_1 \cup P_2) \leq U(P_2, f)$.

Thus

$L(P_1, f) \leq U(P_1, f)$ for any partitions P_1 and P_2.

Hence: $\int_{-a}^{b} f = \sup L(P_1, f) \leq U(P_2, f)$,

for any partition. Taking the infimum over P_2 gives the result.

5.4.1 *Proof 4*

Since f is integrable, for each $\in > 0$, $\exists\ P \in \mathbb{P}$ [a, b] such that $U(P, f) - L(P, f) < \in$.

For this partition $P \in \mathbb{P}$ [a, b] we have $M_i^+ = M_i$ and $M_i^+ \geq m_i$,

where $M_i^+ = \sup\{f^+(x) : x_{i-1} \leq x \leq x_i\}$.

Hence $U(P, f^+) = U(P, f)$ and $L(P, f^+) \geq L(P, f)$,

Thus $U(P, f^+) - L(P, f^+) \leq U(P, f) - L(P, f) < \in$.

Hence f^+ is also integrable. But $f^- = f^+ - f$ is also integrable.

Hence $|f|$ is integrable.

Moreover, $f \leq |f| \Rightarrow \int f dx \leq \int f dx$

Also $-f \leq |f| \Longrightarrow -\int f dx \leq \int f dx$

That is $|\int f dx| \leq \int |f| dx$

5.5.1 *Proof*

Let f: [a, b] $\rightarrow \mathbb{R}$ be continuous. Since f acts on a closed and bounded set (and hence a compact set), then f is uniformly continuous. Thus given $\in > 0$ we can choose a number η such that η(b − a) < $\in$. Then $\exists$ δ > 0 such that for any pair of points x and x',

$|x - x'| < \delta \Rightarrow |f(x) - f(x')| < \eta$

We also have

$|x_i - x_{i-1}| < \delta \Longrightarrow |f(x_i) - f(x_{i-1})| < \eta$

Choosing a partition P of [a, b] such that μ(P) < δ,

$M_i = f(x_i)$, $m_i = f(x_{i-1})$,

we have μ(P) < δ ⇒ $M_i - m_i < \eta$ by (*) .

Thus $U(P, f) - L(P, f) = \sum_{i=1}^{n} M_i \Delta x - \sum_{i=1}^{n} m_i \Delta x$

$= \sum_{i=1}^{n} (M_i - m_i)\Delta x_i < \sum_{i=1}^{n} \eta \Delta x_i$

$= \eta \sum_{i=1}^{n} \Delta x_i = \eta (b - a) < \in$.

Hence f is Riemann integrable.

5.6.0 Examples

5.6.1 Example 1

Lets define partition P of [2,6] such that μ(P),the mesh of P,is small.A possible choice for such a partition could be used to use the power of *e.*

So the partition P can be expressed as:

Now, partition P can be expressed as:

$P = \{[e^2, e^3,], [e^3, e^4], [e^5, e^6]\}$

Let's examine μ(P): Using powers of *e:*

$P = \{[e^k, e^{k+1},], [k \in \mathbb{Z}, 2 \leq e^k \leq e^6]$

Let's examine μ(P):

$\mu(P) = \max\{x_{i+1} - x_i\} = e^{k+1} - e^k = e^k.(e - 1)$

From above the partition *p*:

$\mu(P) = e \cdot (e - 1)$

As *e* is an irrational number,$\mu(P)$ will be approximately 3.195.This means that the mesh of the partition is relatively small,and as $f(x) = e^x$ is monotonic increasing the function will vary only slightly over each subinterval.

5.6.2 Example 2

Let's consider two simple functions defined on the interval [1,4]:

$f(x) = x$

$g(x) = 2 - x$

Both f and g are continuous functions on [1,4] and therefore,Riemann integrable

Let's consider their sum $f + g$ is a constant function,which is also continuous everywhere, and thus Riemann integrable

In this instance, I have shown that the sum of two Riemann integrable functions, $f + g$ is again a Riemann integrable function.

5.6.3 Example 3

For any function, f that is bounded and integrable on [0, 2]. This means that f has a finite maximum and minimum value on the interval, and that the lower and upper sums converge to the same limit as the norm of the partition approaches zero.The norm of a partition is the length of the longest subinterval.The intuition behind the inequality is that adding more points to a partition makes the subintervals.

5.6.4 Example 4

If f is bounded function on [2,4] and f is Riemann integrable,then the absolute value $|f|$ is also Riemann integrable.

Lets consider the Riemann integrability criterion.A function $h(x)$ is Riemann integrable on [a,b] if and only if it is bounded on [a,b] and the set of discontinuities of $h(x)$ has lebesgue measure zero.

Since f is bounded on [2,4],$|f|$ is also bounded on [2,4] because $|f(x)| \leq M$ for some constant *M* since f is bounded.

The absolute value function $|x|$ is continuous everywhere so $|f(x)|$ is also continuous wherever $f(x)$ is continuous,and its discontinuous at the points where $f(x)$ is discontinuous.

Since the set of discontinuities of f has Lebesgue measure zero (as f is Riemann integrable),the set of discontinuities of $|f|$ will also have lebesgue measure zero.Hence $|f|$ satisfies the conditions for Riemann integrability and it is also Riemann integrable on [2,4].

5.6.5 Example 5

Consider the function

Consider the function

f : [0, 2] $\rightarrow \mathbb{R}$ defined by $f(x) = sin^2 x$

We note that f is continuous on [0,$\frac{\Pi}{2}$] and hence it is Riemann integrable on [0,$\frac{\Pi}{2}$] Indeed

$$\int_0^{\frac{\Pi}{2}} sin^2 x dx = \int_0^{\frac{\Pi}{2}} (\frac{1}{2} - \frac{1}{2} cos^2 x) dx$$

$$= \frac{1}{2}x - \frac{1}{4} sin2x \Big|_0^{\frac{\Pi}{2}}$$

$$= \frac{\Pi}{4} - \frac{1}{4} sin\Pi = \frac{1}{4}$$

CHAPTER FOUR

6.0 Conclusions

The **first chapter** of this term paper focuses on defining the key terms and abbreviations related to Riemann integral.It begins by introducing the abbreviations used throughout the paper, such as " =" "to denote equivalent to"a ϵ b" explained to mean that "a" is an element of set "b"," ϕ" denotes an empty set","$\leq$ $\geq$" " To denote less than or equal to and greater than or equal to respectively,"$\forall$" meaning for all,"A$\subseteq$B" signifies that "A" is a subset of "B".

Also literature review is discussed in this chapter where the chapter provides a historical review of the development of Riemann integral.It highlights key contributions from mathematicians throughout the centuries, such as Newton and Leibniz who independently developed the concept of integration in the late 17th century.In 1854 Riemann introduced the Riemann integral on the representation of a function by trigonometric series.In this chapter also the importance of Riemann integral in mathematics society is discussed and also how it has been improved.

Chapter 2 the term paper discusses two types of integral functions where integrable functions and non integrable functions are discussed.Here theorems and examples have been highlighted respectively according to the type of integrable function.

Chapter 3 Riemann integral is defined fully.Concepts of the Riemann integral such as:*partition,norm and tagged partition* have been highlighted and discussed respectively.Riemann sum is also discussed without forgetting properties of Riemann integral such as : *linearity,monotonicity and additivity.*Riemann integral theorems,proofs and examples have been discussed respectively.

Finally,chapter four gives an overview of what is being covered in the above chapters and concludes by providing references.

6.1 References

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